



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KATHIRESWAR & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

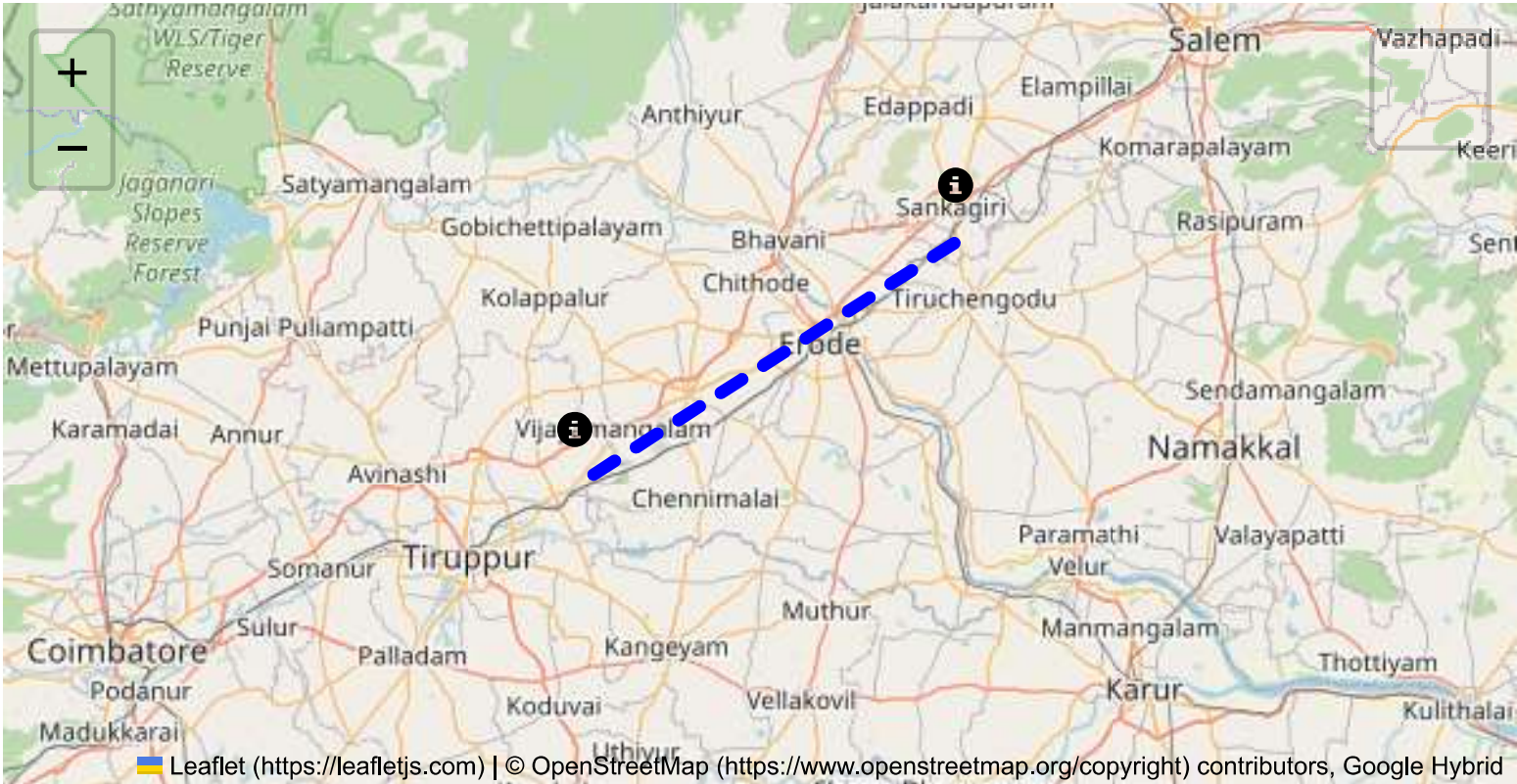
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

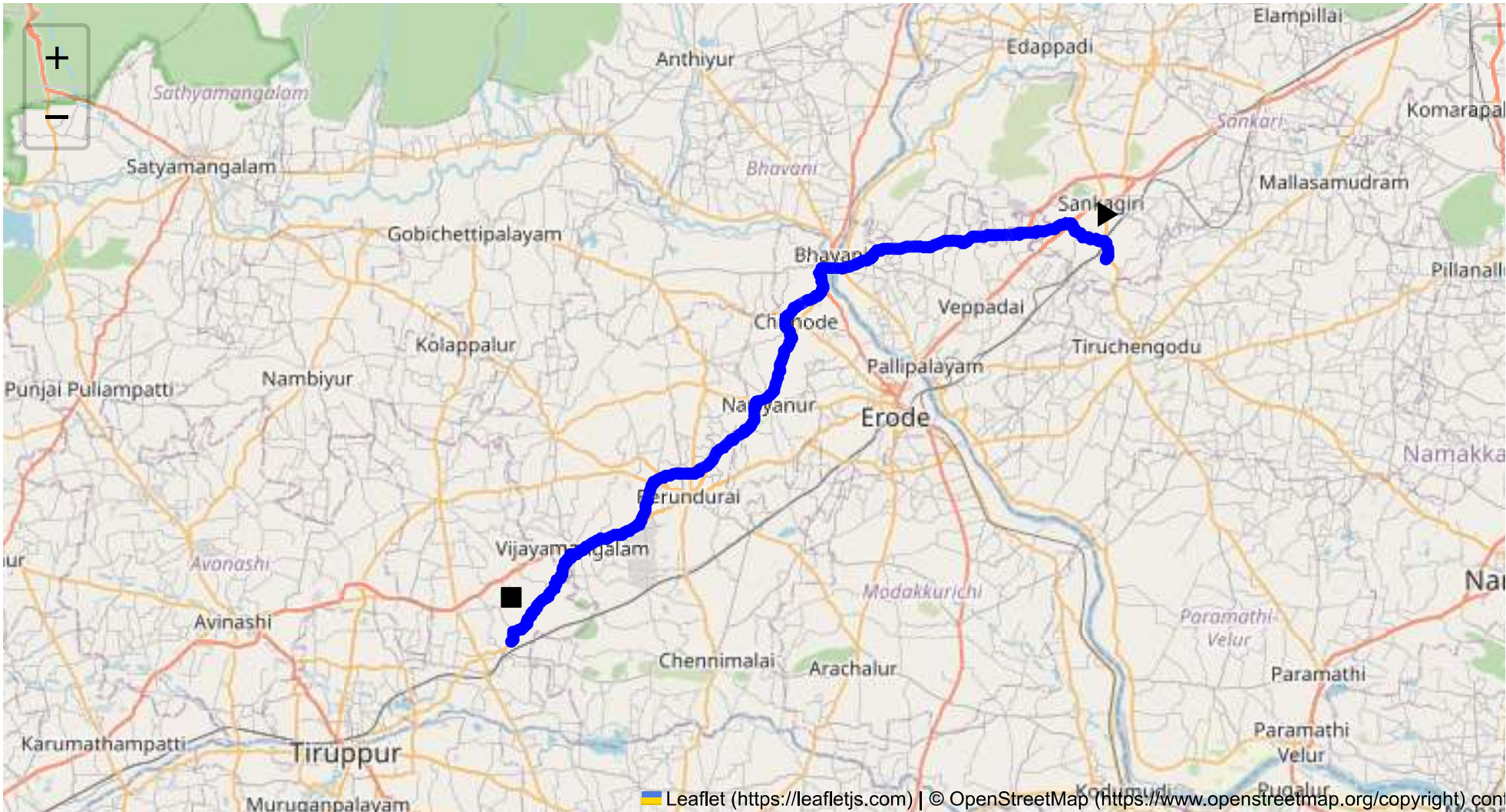
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 65.51 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.176452, 77.459679)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map:

The route from CVQF+23W, Sangagiri, to Dalavoipalayam, Uthukuli, spans approximately 65.51 kilometers. The journey primarily involves navigating through regional highways and rural roads, consisting of varying

traffic conditions and road qualities. The main highways used in this route are NH544 and NH381, known for connecting various industrial areas.

2. Typical Weather Conditions and Potential Hazards:

The region experiences a tropical climate with a hot and humid atmosphere, particularly during the summer months (March to June). The monsoon season (June to September) brings heavy rains, leading to potential road flooding and reduced visibility. Fog can be a hazard in the winter months, affecting visibility in the early mornings.

3. Traffic Patterns and Peak Hours:

The route involves several high-traffic areas, especially near towns like Sankari and Bhavani. Peak travel times are typically between 8:00-10:00 AM and 5:00-7:00 PM, with significant congestion expected during these periods. Traffic is heavier near industrial zones and market areas. The NH544 can also witness periodic congestion due to its role as a major transport corridor.

4. Assessment of Road Quality and Infrastructure:

The road quality varies throughout the route:

- **NH544:** Generally well-maintained, but can suffer from potholes post-monsoon.
- **Rural Roads:** The quality can be inconsistent with occasional rough patches, particularly in less urbanized areas.
- Bridges and overpasses should be approached with caution, checking for weight capacity clearance.

5. Suggestions for Alternative Routes for Emergencies:

In case of roadblocks or emergencies, an alternative route includes diverting through Erode - Perundurai - Kangeyam - Uthukuli, though this adds approximately 20 km to the journey. It helps bypass heavily congested areas.

6. Local Regulations Affecting Hazardous Material Transport:

Trucks carrying hazardous materials must adhere to restrictions, such as avoiding travel during rush hours and conforming to speed limits (usually 40-50 km/h within populated zones). Special permits may be required for certain categories of hazardous goods.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

The region has witnessed incidents involving tanker rollovers and spillage, primarily on NH544 near Bhavani. These were exacerbated by speeding and narrow turns. Safety audits frequently emphasize improving driver training and road conditions.

8. Environmental Considerations and Sensitive Areas:

The route passes through agricultural zones, where chemical spills could severely impact soil and water quality. Awareness of local flora and fauna crossings is necessary, especially in more rural segments.

9. Communication Coverage and Potential Dead Zones:

Overall communication coverage via mobile networks is strong along the NH highways, but dead zones may occur in rural stretches near Bhavani and Sankari. It is advisable to have offline maps and emergency contacts available.

10. Estimated Emergency Response Times:

- **Highway Sections:** Approximately 20-30 minutes for emergency services to respond.
- **Rural Roads:** Can extend beyond 45 minutes due to difficult access and lower service availability.

12. Overall Summary of Risk Assessment:

The route entails moderate risk, particularly during inclement weather conditions and peak traffic hours. Key hazards include variable road quality and potential traffic congestion. Preparedness for communication disruptions and knowledge of local regulations are critical for safely transporting hazardous materials. Alternative routes and emergency response plans should be pre-determined to mitigate potential delays or incidents. Enhanced driver awareness and training can alleviate the impact of sudden hazardous situations.

In conclusion, thorough preparation, adherence to regulations, and acute awareness of environmental and infrastructural challenges are key to ensuring a safe journey along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44898, 77.87410	30 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	Medium	11.45509, 77.81383	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.29288, 77.59029	30 KM/Hr
11	Turn	Medium	11.29140, 77.58253	30 KM/Hr
12	Turn	Medium	11.29154, 77.58236	30 KM/Hr
13	Blind Spot	Blind Spot	11.23716, 77.50666	10 KM/Hr
14	Blind Spot	Blind Spot	11.23704, 77.50676	10 KM/Hr
15	Blind Spot	Blind Spot	11.23686, 77.50216	10 KM/Hr
16	Turn	Medium	11.23649, 77.50219	30 KM/Hr
17	Turn	Medium	11.23396, 77.50105	30 KM/Hr
18	Turn	Medium	11.23329, 77.49966	30 KM/Hr
19	Turn	Medium	11.17851, 77.46169	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
8	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
9	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
10	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium
11	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
12	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
13	clinic	Erode Cancer Centre	11.3732, 77.649152	30 km/h	Medium
15	hospital	Irt Hospital	11.2803603, 77.5644118	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Gen Siddha Hospital	11.2375033, 77.5059797	30 km/h	Medium
18	hospital	Vijayamangalm Government Hospital	11.2382827, 77.501687	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
7	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
14	college	Government polytechnic college	11.2907053, 77.5698726	30 km/h	Medium
16	school	Bharathi Matriculation School	11.2494613, 77.5307028	30 km/h	Medium
19	college	Sasurie Engineering College	11.2171482, 77.4898552	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45509, 77.81383



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.29288, 77.59029



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.29140, 77.58253



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.29154, 77.58236



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23716, 77.50666



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23704, 77.50676



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23686, 77.50216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23649, 77.50219



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23396, 77.50105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23329, 77.49966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17851, 77.46169

